Matthew McDonald

Undergraduate Biology Student
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EDUCATION

College of Charleston, Charleston SC — *B.S Biology*

Advisor: Daniel McGlinn

GPA: 3.89/4.00

Expected Graduation: May 2026

RESEARCH INTERESTS

- Effects of habitat loss, fragmentation, and habitat management on herpetofauna at population and community levels
- Reptile and amphibian community composition changes in response to climate change
- Natural history of understudied species
- Biogeography and Phylogeography

FUNDING AND ACADEMIC AWARDS

- **Research Grants and Stipends**
  - 2025 College of Charleston SNES Undergraduate Research Grant (6,000)
  - 2023 College of Charleston SSME Undergraduate Research Grant (6,000)
- **Scholarships**
  - 2023 - Present SC LIFE Scholarship and STEM Enhancement (7,500/year)
  - 2025 National Association of Environmental Professionals SC Scholarship (2,000)
  - 2025 Harry W. Freeman Endowed Scholarship For Biology (1,450)
  - 2025 Garden Club Scholarship for Scientific and Community Contributions to Plant Sciences (1,500)
  - 2024 College of Charleston Maggie T. Pennington Scholarship for Excellence in Biology (3,000)
  - 2022 SC HOPE Scholarship (2,800)

- **Travel and Conference Awards**
    - 2024 CIE Travel Abroad Scholarship (2,000)
    - 2024 SACNAS Conference National Travel Scholarship (600)

RESEARCH AND FIELD EXPERIENCE

Independent Undergraduate Research, Charleston — *Undergraduate Researcher/Research Assistant*

2022 - Present

- Funded research to assess the impact of pine savannah management regimes on the population ecology of arthropods in seasonal wetlands. Responsibilities include:
    - Field work such as installing and monitoring insect pitfall traps and sweep netting
    - Identifying over 4400 arthropods to the order level, and recording abundance and biomass data
    - Generating protocols, manuscripts, budgets, and other research documentation
    - Analyzing community metrics such as diversity and abundance in Rstudio
    - Disseminating results to the peers and public
- Assisted graduate and undergraduate research projects in ecological fieldwork:
    - Conducted reptile and amphibian surveys using pitfall traps, artificial cover, and visual encounter surveys
    - Conducted bird diversity surveys in ephemeral wetlands
    - Assisted in habitat management, prescribed burns, and conservation efforts for the endangered Canby's Dropwort plant in ephemeral wetlands
    - Conducted wetland vegetation identification and monitoring

REU Research, California Academy of Sciences — *Internship*

2024

- Competitive research internship (acceptance rate ~ 2%). 7 week full time paid internship at the California Academy of Sciences Summer Systematics Institute. Investigated the phylogeography of the Baja Coachwhip Snake using low coverage whole genome sequencing of mitochondrial DNA. Responsibilities include:
    - Creating programs within the Bash shell programming language to sequence mitochondrial DNA using Low Coverage Whole Genome Sequencing for over 70 mitochondrial genomes
    - Producing phylogenies using IQtree and GENEious programs
    - Writing manuscripts, and presenting results to researchers and public

SCDNR, Charleston — *Wildlife Technician*

2021 - 2022

- Preparing and assisting in installing recycled oyster reefs and Spartina Grass as a means of habitat restoration and erosion mitigation. Responsibilities include:
    - Installing oyster reefs and salt marsh grass in community based restoration efforts

• Surveying reef biodiversity through gill and seine netting

• Collecting geospatial data through Trimble GIS

• Growing and planting of Spartina marsh grass

• Educating students and the public on coastal natural resources. Assisted in outreach with over 2,000 community members and students

• Towing and operation of medium-sized boats in coastal SC waters

Amphibian and Reptile Conservancy , South Carolina — *Volunteer*

2020 - Present

•Volunteer for field work with the nonprofit Amphibian and Reptile Conservancy. Responsibilities include:

• Radio Telemetry tracking of Eastern Diamondback Rattlesnakes and Northern Pine Snakes in the Francis Marion National Forest

• Monitoring of drift fences and aquatic traps for snakes and state threatened Spotted Turtles

• Assisting in habitat enhancement in ephemeral wetlands for the state endangered Gopher Frog

Turtle Survival Alliance , South Carolina/ Florida — *Volunteer*

2020 - 2022

• Volunteer in the two-day population survey of Weeki Wachee Springs State Park in Florida. Species surveyed included species in the *Pseudemys* and *Trachemys* genera. Responsibilities include:

• hand capture of turtles by snorkeling

• weighing, measuring, marking, and assisting in PIT tagging. Over 100 individuals recorded

• Assisting in 2019-2022 population surveys on Kiawah Island SC for Diamondback Terrapins. Surveying responsibilities include:

• visual surveying and seine netting of intertidal creeks in and around the Kiawah River. Over 40 individuals recorded

Alliance for Planet Earth Club, South Carolina — *Club Leader*

2022 - 2025

• Organized and led field trips for over 300 students to College of Charleston's 1,000 acre Stono Preserve to assist in habitat management and to teach students about local ecology. Responsibilities include:

• Leading efforts in habitat restoration such as planting native trees, removing invasive plant species, and installing bird nesting boxes

• Leading guided tours to educate students on local natural history

• Coordinating with land managers for monthly trips to College of Charleston's Stono Preserve

Various Organizations, South Carolina — *Volunteer*

2020 - Present

• Installation of habitat barriers for head started Gopher Tortoises - SCDNR, Aiken Gopher Tortoise Preserve

• Assisting in prescribed burns of Longleaf Pine Savannah - Audubon Society

• Assisting in habitat enhancement for Black Rails in wetlands - SCDNR

• Assisted with outreach to over 100 high school students at Wando High School with live reptiles and amphibians to promote conservation awareness and environmental education - Wando High School

MANUSCRIPTS IN PREPARATION

·**MC McDonald**, D.J McGlinn *Ephemeral Wetlands and Arthropod Diversity*

·**MC McDonald**, *Natural History Note: Relictual Spotted Turtle Populations in Fragmented Suburban Wetlands*

CONFERENCES AND PRESENTATIONS

· **Conference Presentations**

· 2025 Poster Presenter, Joint Meeting of Ichthyologists and Herpetologists, *Phylogeographic Patterns in the Baja Coachwhip (Masticophis fuliginosus)*

· 2024 Poster Presenter, Sea Grant Consortium Forested Wetlands Conference, *Ephemeral Wetlands are Not Biodiversity Hotspots For Arthropods*

· 2024 Poster Presenter, SACNAS Diversity in STEM Conference, *Phylogeographic Patterns in the Baja Coachwhip (Masticophis fuliginosus)*

· 2024 Speaker, REU Research Symposium, *Phylogeographic Patterns in the Baja Coachwhip (Masticophis fuliginosus)*

· 2024 Poster presenter, Undergraduate Research Symposium, College of Charleston, *Phylogeographic Patterns in the Baja Coachwhip (Masticophis fuliginosus)*

· 2023 Poster presenter, Undergraduate Research Symposium, College of Charleston, *Ephemeral Wetlands are Not Biodiversity Hotspots For Arthropods*

· **Other Presentations**

· 2025 Invited Speaker, College of Charleston Darwin Week Talks, *Roadkill Genetics: Insights into Evolution from Dead Snakes*

· 2024 Guest Lecturer, *How To Restore Isolated Wetlands*, College of Charleston, Introduction to Environmental Sustainability Class

· 2023 Invited Speaker, *Ecology of Wetland Arthropods*, College of Charleston Biology Club

HIGHLIGHTED COURSES

- Herpetology, Conservation Biology, Ecology (Study Abroad in Panama), Entomology (Fall 2025), Ichthyology, Genetics, Statistics (emphasis in Rstudio) (Fall 2025), Geospatial Science (emphasis in ARCGIS) (Fall 2025), Environmental Geology

REFERENCES

Dan McGlinn, Ph.D- Associate Professor of Biology — College of Charleston
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Edward Meyers, Ph.D. - Research Scientist at the California Academy of Sciences
Email: emyers@calacademy.org

Michael Hodges, Head Biologist — SCDNR Oyster Recycling and Enhancement Program
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